

Fig. 3: An actual-size, single-side PCB for PIR security light

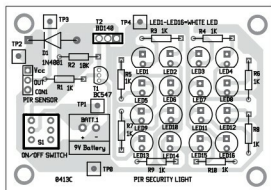


Fig. 4: Component layout for the PCB

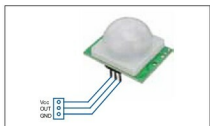


Fig. 5: Pin configuration of PIR sensor

T2 (BD140). Now the white LEDs are powered through T2 for a finite duration, determined by the time delay setting of the PIR sensor module.

Note that the PIR sensor has a 'time' control, which is adjustable from seconds to minutes. When infrared radiation from a moving heat source is detected, the sensor output goes high and begins timing out as per the preset time period. After the pre-set time, the output goes low.

According to the data sheets, PIR sensor requires an initial stabilisation time of 10 to 60 seconds in order to function properly. During this time, the sensor gets familiar with the sur-

rounding environment, and any motion in its field of view should be avoided. Typically, the PIR sensor has a range of 6 metres (20 feet), and is designed to adjust to slowly changing conditions such as gradual change in the thermal profile of the surroundings as the day passes.

Construction and testing

An actual-size, single-side PCB for PIR security light is shown in Fig. 3 and its component layout in Fig. 4. Before assembly, refer to Fig. 5 and the datasheet of PIR sensor in order to connect the sensor with correct pin configurations. After assembling the circuit on a PCB, enclose it in a suitable case. Fix the PIR sensor on the PCB where a suitable connector (CON1) is provided.

Check 9V at test point TP1 with respect to TP0 to verify the correct power supply to the circuit. Turn switch S1 on and measure the voltage at test point TP2 with respect to TP0, it should be 0.7V (diode voltage) less than the voltage at TP1. Activate the sensor and measure the voltage at TP3, which should be 3.3V and, at the same time, voltage at TP4 should be approximately 9V for the LEDs to glow.

EFY note. 1. In some PIR sensors (like SB0081), a fourth pin (labelled as CDS) is available, which can be used to add an optional light-dependent resistor (CdS photocell) for ambient light detection.

2. The sensor responds to any sudden change in the thermal profile. That's why the PIR sensor module should not be placed near a heater, AC outlet or anything that could cause a rapid change in the temperature of surrounding environment. ●

The author is a regular contributor to EFY and a freelance writer



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